Multimodal IoT Predictive Maintenance (PdM) Blueprint

Target System: Samsung WW10TP44DSX/FQ 10kg AI Ecobubble Washer | **Academic Context:** FYP Multimodal PdM

1. Complete Bill of Materials (BOM) & Purchasing List

#	Component Name	Shop & Variant	Price	Hardware Role
1	**NodeMCU ESP32 (30-Pin)**	Robotedu.my ESP32 Board + USB	RM 32.90	240MHz dual-core MCU, Wi-Fi/BLE, hardware I2C & I2S DMA controllers.
2	**ESP32 Expansion Shield**	MakerHub MY ESP32 EXPANSION BRD (30P)	RM 8.46	Screw terminal interface. Clamps wires to survive 1400 RPM spin vibration.
3	**MPU-6050 Accelerometer**	ElectricA Ecom Soldered Headers	RM 21.99	3-axis vibration/acceleration sensing (configurable up to ±16g @ 200Hz).
4	**INMP441 MEMS Microphone**	ElectricA Ecom Soldered Headers	RM 25.99	Digital 24-bit I2S acoustic sensor for mechanical audio emissions.
5	**Dupont Wires (40pcs)**	MakerHub MY M-F 20cm	RM 3.20	Connects sensor pins (F) directly to screw shield terminals (M).
6	**ABS Waterproof Project Box**	baba168.my 158x90x60mm	RM 16.30	IP65 weatherproof housing; flat base provides solid structural coupling.
7	**PG7 Cable Glands (5pcs)**	CASSETECH WHITE PG7	RM 2.50	Watertight rubber compression seal for USB cable entry.
8	**3M Original VHB Tape**	3M shop 2.5cm x 3m (0.8mm)	RM 14.61	High-stiffness acrylic bond. Transmits vibration without damping signals.
	ESTIMATED TOTAL	All Components Complete	**~RM 125.95**	Ready-to-deploy multimodal hardware testbed.

2. Pin-by-Pin Wiring Diagram & Pinout Table

Both sensors are powered via the **3.3V (3V3)** rail. The MPU-6050 communicates via **I2C**, while the INMP441 streams audio over **I2S DMA**.

Sensor	Sensor Pin	ESP32 Pin	Protocol / Signal	Wire Color	Functional Description
MPU-6050	**VCC**	**3V3**	Power (3.3V)	Red	Clean regulated logic power supply
MPU-6050	**GND**	**GND**	Ground (0V)	Black	Common ground return path
MPU-6050	**SCL**	**GPIO 22**	I2C Clock	Yellow	I2C Serial Clock synchronization line
MPU-6050	**SDA**	**GPIO 21**	I2C Data	Green	Bi-directional I2C telemetry data line
MPU-6050	**AD0**	**GND**	Address Pin	Blue	Ties address bit to lock I2C address to 0x68
INMP441	**VDD**	**3V3**	Power (3.3V)	Red	Digital MEMS microphone power supply
INMP441	**GND**	**GND**	Ground (0V)	Black	Common audio ground
INMP441	**SCK**	**GPIO 33**	I2S BCLK	Orange	Bit Clock for continuous serial audio bits
INMP441	**WS**	**GPIO 25**	I2S LRCLK	White	Word Select / Left-Right Frame clock
INMP441	**SD**	**GPIO 32**	I2S DIN	Gray	Serial Data input stream into ESP32 DMA
INMP441	**L/R**	**GND**	Channel Sel	Purple	Pull to GND to stream mono on Left channel

3. Step-by-Step Checklist: What to Do Once Everything Arrives

Phase 1: Benchtop Testing & Smoke Verification (30 Minutes)

- **Seat the ESP32:** Align the 30-pin board into the MakerHub expansion shield. Inspect all pins to ensure none are bent.
- **Plug in USB:** Connect to PC and verify COM port detection under Device Manager (CP2102 driver).
- **I2C Verification:** Wire MPU-6050 (VCC, GND, GPIO 21, GPIO 22, AD0->GND). Run an I2C Scanner sketch in Arduino IDE. Verify device acknowledges at address 0x68.
- **I2S Acoustic Verification:** Wire INMP441 (GPIO 33, 25, 32). Flash a test script and view live audio waveforms in the Arduino Serial Plotter.

Phase 2: Enclosure Preparation & Mounting (45 Minutes)

- **Cable Gland Hole:** Drill a 12mm hole on the side/bottom wall of the ABS box. Fit the PG7 gland with its rubber O-ring.
- **Acoustic Port:** Drill a 1.5mm - 2mm small pinhole facing downward for the INMP441 microphone diaphragm.
- **Internal Sensor Anchor:** Tape the MPU-6050 flat against the interior base of the ABS box with a small piece of VHB tape. (Never allow the accelerometer to float loose on jumper wires).
- **Chassis Bonding:** Clean the top-rear metal casing of your Samsung washer with alcohol. Stick the enclosure down with 3M VHB tape and apply firm pressure for 30 seconds.

Phase 3: Machine Baselines & Synthetic Fault Experiments

- **Healthy Baseline Run:** Record telemetry during a standard Daily/Cotton cycle (3-4kg mixed clothing) at 1400 RPM.
- **Synthetic Fault 1 (Unbalance):** Run a spin cycle with 1 single heavy wet bath towel to test the 1X RPM unbalance anomaly detector.
- **Synthetic Fault 2 (Loose Foot):** Loosen one front rubber leveling foot by 2 turns to test rocking and structural harmonic anomalies.